

Name: _____

Student ID: _____

Signature: _____

UNSW Sydney
TERM 1 2026 MIDTERM EXAMINATION
ELEC3115: Electromagnetic Engineering

1. TIME ALLOWED – 1 hour

2. READING TIME – 10 minutes

3. THIS EXAMINATION PAPER HAS 3 PAGES

4. TOTAL NUMBER OF QUESTIONS – 4

(Approximate time guide (not strict): Q1: 25 min | Q2: 20 min | Q3: 10 min | Q4: 5 min

5. TOTAL MARKS AVAILABLE – 50

6. MARKS AVAILABLE FOR EACH QUESTION ARE SHOWN IN THE EXAMINATION PAPER

7. ALL ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY BE USED ONLY FOR DRAWING, SKETCHING OR GRAPHICAL WORK

8. THIS PAPER MAY **NOT** BE RETAINED BY CANDIDATE

9. CANDIDATES MAY BRING TO THE EXAMINATION an UNSW-approved calculator and A4-formual page (both sides are permitted to use – handwritten or printed). The formula page must be returned along with the question paper and answer booklet.

10. Any assumptions found to be necessary in answering the questions should be stated explicitly. Must show all your workings to receive any partial marks applicable.

Question 1 [20 marks]

- (a) [12 marks] A very long transmission line is located above the ground at a height h m. Consider that the influence of the ground plane is negligible at this height. The line is charged with a uniform line charge density ρ_l C/m.
- (i) [5 marks] Starting from Gauss' law, derive an expression for the electric field at a radial distance r from the centre of the line.
- (ii) [5 marks] Using the electric field obtained in part (i), derive an expression for the potential difference between two radial distances r_1 and r_2 .
- (iii) [2 marks] Briefly explain why the influence of the ground plane would invalidate the derivation in parts (i) and (ii).
- (b) [5 marks] The transmission line has a diameter of 1.2 cm and is located 7 m above the ground. Calculate the potential difference between the surface of the line and a reference point at the ground below. The line charge density is 100 nC/m. Given $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.
- (c) [3 marks] Briefly describe why we need to use the '*method of image*' to calculate the capacitance between the transmission line and the ground

Question 2 [16 marks]

- (a) [4 marks] Explain the difference between polarization (physical process) and polarization vector $\vec{P}$ (mathematical quantity) in a dielectric material?
- (b) [8 marks] A parallel-plate capacitor of 0.3 m^2 area and plate separation of 10 mm is filled with a dielectric material with relative permittivity $\epsilon_r = 3$. The top and the bottom plates have charges of +0.4 and -0.4 micro-Coulombs, respectively. Assuming no fringing of the electric field, calculate the polarization vector $\vec{P}$ in the material.
- (c) [4 marks] If the capacitor is filled with two stacked dielectric slabs with the upper slab thickness of 6 mm and $\epsilon_r = 3$, and the lower slab thickness of 4 mm and $\epsilon_r = 6$, show using boundary (or interface) conditions for the normal components of the electric flux density $\vec{D}$ that the electric field in the upper slab is two times stronger than that of the lower slab.

Question 3 [10 marks]

(a) [8 marks] Capacitance formation in a 3-conductor transmission system is shown in Figure 1. V_1 , V_2 , and V_3 are the potential of the conductors. Capacitance values obtained from the measurement are shown in the figure. Use the given capacitance values to construct the capacitance matrix for this network, which relates the charges and potentials of the conductors. Clearly label your matrix elements.

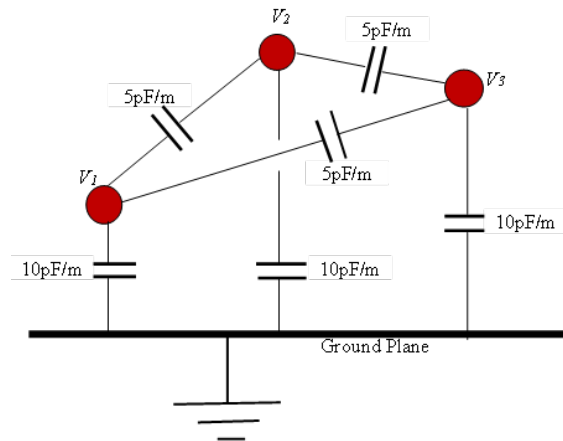


Figure 1

(b) [2 marks] List an advantage that the capacitance matrix brings as a calculation tool in a capacitive network of a multi-conductor system.

Question 4 [4 marks]

[4 marks] State the two fundamental observations about the static magnetic field that lead to the following fundamental postulates:

1. $\vec{\nabla} \cdot \vec{B} = 0$

2., $\vec{\nabla} \times \vec{B} = \mu \vec{J}$

END OF EXAM PAPER